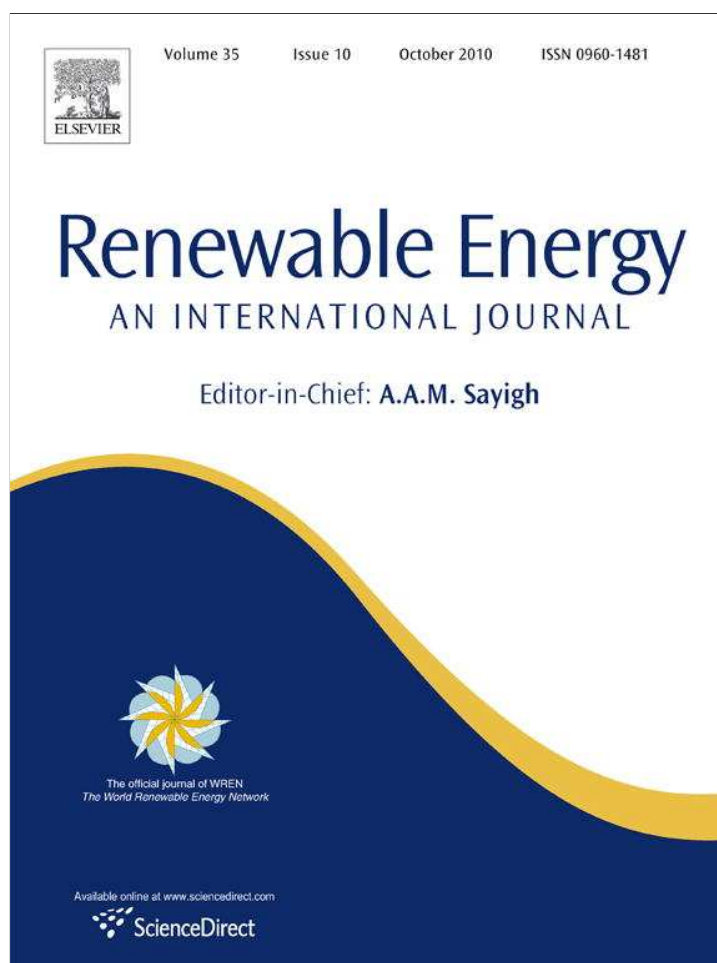




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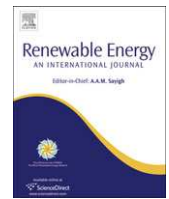
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Short-term prediction of wind power with a clustering approach

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ABSTRACT

A clustering approach is presented for short-term prediction of power produced by a wind turbine at low wind speeds. Increased prediction accuracy of wind power to be produced at future time periods is often bounded by the prediction model complexity and computational time involved. In this paper, a trade-off between the two conflicting objectives is addressed. First, a set of the most relevant parameters (predictors) is selected using the underlying physics and pattern immersed in data. Five scenarios of the input space are created with the *k*-means clustering algorithm. The most promising clustering scenario is applied to produce a model for each clustered subspace. Computational results are compared and the benefits of cluster-specific (customized) models are discussed. The results show that the prediction accuracy is improved the input space is clustered and customized prediction models are developed.

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1. Introduction

Knowing the power to be produced by a wind turbine at future time horizons is of interest to the rapidly expanding wind industry [1]. Wind power forecasts are used as input for various tools, e.g., management of power dispatch and control of wind turbines [2,3].

The state-of-the-art approaches to wind power forecasting have been published in [4] with more recent updates included in [5]. Models used for forecasting wind power are categorized as physics-based models, statistical models, and spatial correlation models [3,6–10]. Data-mining algorithms offer a promise to conquer the unresolved gap of handling the dynamic nature of wind [11].

The published literature on data-mining in wind power is growing, with Neural Networks (NNs) becoming the widely used algorithm. NN models can be used to estimate power output as a function of wind turbine parameters (e.g., wind speed, generator torque) and time delay of the corresponding parameters (e.g., power itself, wind speed) [12,13]. Wind speed, relative humidity, and time were used as input variables to train an NN model in power prediction applications [14,15]. The recurrent multilayer-perceptron NN was applied for power prediction in [16]. Long- and short-term prediction of power using the *k*-nearest neighbor (*k*-NN) algorithm was presented in [11,12]. Analysis and estimation of power based on cluster analysis was reported in [17].

This paper is organized in eight sections. Section 2 describes the data used for this paper. The candidate parameters of interests to

the studied domain are selected using known equations. Section 3 discusses the proposed clustering approach (model customization) for short-term power prediction. Parameter selection by data-mining algorithms is presented in Section 4. Section 5 discusses and compares five scenarios of clustering the input space. Section 6 presents models extracted from data of each clustered subspace. Comparison between the proposed model and the model extracted in typical ways is discussed in Section 7. Section 8 concludes the paper.

2. Data description and parameter selection

Wind turbine data is usually collected by a Supervisory Control and Data Acquisition (SCADA) system. Though the data sampling frequency may be relatively high (e.g., 20 Hz), the data is averaged into time intervals, e.g., 10 s, 30 s, or 10 min, that are suitable for various applications. The data used in this paper was collected at 10 s intervals (called 10 s data) at a 1.5 MW wind turbine (randomly selected) for a period of seven days. For the selected wind turbine, the cut-in speed is 3.5 m/s, the rated speed is 12.5 m/s, and cut-out speed is 21 m/s. From the view of turbine operations, wind speed in the range [3.5 m/s, 12.5 m/s] is of interest to industry. Thus, the data with a wind speed lower than 3.5 m/s or higher than 12.5 m/s have been excluded from analysis in the research reported in this paper. Data points with a negative power output have been also deleted. The data from the first five days (approximately 2/3 of all data) was used to extract models, and the data from the remaining two days (approximately 1/3 of all data) was used for test and validation models. The data set used in this research is characterized in Table 1.

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Table 1

Description of the training and test data.

Data set	Start time	End time	Time interval	Number of data points
Training	8/8/07 12:00 AM	8/12/07 12:00 AM	10 s	30 354
Test	8/13/07 12:00 AM	8/15/07 12:00 AM	10 s	15 860

The data available for this research included numerous parameters of a wind turbine. Some of these parameters could have a potential impact on the prediction accuracy of wind power to be generated at 10 s intervals.

Based on the basic wind power equation [18], five parameters are selected as candidates, including wind speed v , blade pitch angle β , generator torque T , and rotor speed ω_r . Considering the fact that the system inertia could be significant, the power output P_a is also included. The air density ρ and rotor radius R are regarded here as constants. The initially selected parameters are listed in Table 2.

Since time delay is considered as having an impact on the model's accuracy, it also considered, and thus further parameter selection is accomplished with data-mining algorithms (see Section 4).

3. Proposed methodology

The input data (controllable, non-controllable, and performance parameters) representing the input space undergo parameter selection and clustering. Based on the data in each cluster, a model is produced with data-mining algorithms. The number of input parameters n (dimension) and the number of instances N define the input space IS_n . For each parameter in the input space, there are $N = 30\,354$ training instances 1 and $N = 15\,860$ test instances (see Table 1). Each of the models $1, \dots, k$ predicts power output $PO(t)$ at time t .

The steps of the proposed methodology are discussed next.

3.1. Parameter selection

The five parameters listed in Table 2 partially describe the input space. To simplify the input space, the same number m of past

Table 2

List of parameters selected for wind speed estimation.

Parameter type	Parameter name	Abbreviation	Symbol	Unit
Non-controllable	Wind speed	WS	v	m/s
Controllable	Blade pitch angle	BPA	x_1	°
	Generator torque	GT	x_2	Nm
Performance	Power output	PO	y_1	kW
	Rotor speed	RS	y_2	rpm

c) Clustering on generator torque $x_2(t)$

d) Clustering on generator torque $x_2(t)$ and rotor speed $y_2(t-1)$

e) Clustering on generator torque $x_2(t)$, rotor speed $y_2(t-1)$, and power output $y_1(t-1)$

The first two scenarios explore the impact of wind speed on the accuracy of power output predictions at 10 s intervals. As the wind speed $v(t)$ at future time t is not known, two estimation methods have been applied. The first one (item (a) above) uses wind speed at one past state $v(t-1)$, and the second one is based on the time series model. Prediction of wind speed with the time series models has been proven to be accurate (see [19]).

The final three scenarios of clustering the input space originate in predictors' importance (see Section 4). The first three most significant parameter states determined by the data-mining algorithms are the generator torque $x_2(t)$, rotor speed $y_2(t-1)$, and power output $y_1(t-1)$ (see Section 4). All parameters are studied for impact on clustering the input space.

3.2.2. Clustering the test data set

For the $ndimensional$ space, the center (centroid) of the i th cluster of the training data is denoted as $[x_1^i, x_2^i, \dots, x_n^i]$, where i is the number of the cluster satisfying $1 \leq i \leq k$. To balance the bias due to the variability of the input data the values of $[x_1^i, x_2^i, \dots, x_n^i]$ have been normalized in the interval $[0,1]$.

Thus, the distance from a normalized instance $[z_1, \text{normalized}, \dots, z_n, \text{normalized}]$ to the i th cluster centroid of the training data is defined in (2).

$$D^i = \sqrt{(z_{1, \text{normalized}} - x_{1, \text{normalized}}^i)^2 + \dots + (z_{n, \text{normalized}} - x_{n, \text{normalized}}^i)^2} \quad (2)$$

states is considered for each of the five parameters. The input space IS_n for the m past states of five parameters listed in Table 2 is defined by vector (1).

$$IS_n = [v(t-1), \dots, v(t-m), x_1(t), \dots, x_1(t-m), x_2(t), \dots, x_2(t-m), y_1(t-1), \dots, y_1(t-m), y_2(t-1), \dots, y_2(t-m)] \quad (1)$$

Of all n predictors, the most significant are selected with a NN algorithm (see Section 4).

3.2. Clustering input space

3.2.1. Clustering training data set

In this section, the training data set is clustered into k subspaces. Based on parameters selected for clustering, the following five data processing scenarios are considered:

- Clustering on wind speed estimated by its one past state
- Clustering on wind speed estimated by the time series model

The aim is to find, for each data instance, a cluster with the minimum distance between the instance and the cluster centroids. The clustering algorithm of the test data set is shown in Table 3.

As illustrated in Table 3, $D^{\min}$ is initially set to 1, and then it is replaced with a shorter distance found. In this way, each instance from the test data set is assigned to the closest cluster.

Table 3

Algorithm for clustering test instances.

```

Begin
For  $i = 1$  to  $k$ 
 $D^{\min} = 1$ 
 $D^i = \sqrt{(z_{1, \text{normalized}} - x_{1, \text{normalized}}^i)^2 + \dots + (z_{n, \text{normalized}} - x_{n, \text{normalized}}^i)^2}$ 
If  $D^i \leq D^{\min}$ 
Let
 $D^{\min} = D^i, [z_{1, \text{normalized}}, \dots, z_{n, \text{normalized}}] = [z_{1, \text{normalized}}^i, \dots, z_{n, \text{normalized}}^i]$ 
Else
Next  $i$ 
End

```

Table 4
The 10 most important predictors of Eq. (5).

Input no.	Parameter state	Importance	Percentage (%)
1	GT(<i>t</i>)	781.48	94.28
2	RS(<i>t</i> − 1)	11.53	1.39
3	RS(<i>t</i> − 2)	3.82	0.46
4	PO(<i>t</i> − 1)	3.30	0.40
5	RS(<i>t</i> − 3)	1.99	0.24
6	GT(<i>t</i> − 1)	1.87	0.23
7	PO(<i>t</i> − 2)	1.86	0.22
8	RS(<i>t</i> − 4)	1.53	0.18
9	PO(<i>t</i> − 3)	1.48	0.18
10	PO(<i>t</i> − 5)	1.48	0.18
Total importance		828.89	97.76

3.2.3. Evaluation of clustering effectiveness

The reason for clustering the input space is to achieve a high prediction accuracy of the customized model with a few input parameters. To demonstrate the effectiveness of clustering, NN models are constructed for five different clustering scenarios. A clustering scenario with the highest prediction accuracy is used for the final model extraction and validation.

3.3. Model extraction and validation

After all instances from the training and test data sets have been assigned to the corresponding clusters, data-mining models are constructed and validated. For each training data set of a given cluster, models are extracted with data-mining algorithms. Every model is tested using the corresponding test data set.

The structure of the cluster-based power prediction model is presented in (3).

$$y_1^*(t) = \begin{cases} 0 & \text{if } v(t) < 3.5 \text{ m/s} \\ f_1(\cdot) & \text{if } IS_n \in \text{Cluster}_1 \\ f_2(\cdot) & \text{if } IS_n \in \text{Cluster}_2 \\ \dots & \dots \\ f_k(\cdot) & \text{if } IS_n \in \text{Cluster}_k \\ 1500 & \text{if } v(t) > 12.5 \text{ m/s} \end{cases} \quad (3)$$

The predicted power output $y_1^*(t)$ is assumed to be 0 when wind speed is less than 3.5 m/s, and it equals 1500 kW for wind speed higher than 12.5 m/s. For the wind speed in [3.5, 12.5] m/s, the input space is clustered into *k* subspaces. The prediction accuracy for the test data in each subspace is discussed in Sections 5 and 6.

4. Parameter selection

Algorithms such as the Boosting Tree Algorithm (BTA), Neural Network (NN), and the Random Forest Algorithm (RFA), can be used to determine the importance of parameters and their past states

Table 5
Ranked importance of predictors of Eq. (6).

No. of Inputs	Parameter State	Importance	Percentage
1	GT(<i>t</i>)	865.32	95.61
2	RS(<i>t</i> − 1)	25.15	2.78
3	PO(<i>t</i> − 1)	2.76	0.31
4	RS(<i>t</i> − 2)	2.61	0.29
5	GT(<i>t</i> − 1)	2.27	0.25
6	PO(<i>t</i> − 2)	1.86	0.21
7	RS(<i>t</i> − 4)	1.50	0.17
8	RS(<i>t</i> − 3)	1.34	0.15
9	PO(<i>t</i> − 3)	1.15	0.13
10	PO(<i>t</i> − 5)	1.07	0.12

Table 6
Importance of predictors of Eq. (7).

No. of Inputs	Predictor	Importance	Percentage
1	GT(<i>t</i>)	824.83	96.14
2	RS(<i>t</i> − 1)	30.48	3.55
3	PO(<i>t</i> − 1)	2.63	0.31

(predictors). As the NN algorithm has been used numerous times in this paper, it is also applied to compute absolute importance. From that, the relative importance of parameters is computed (4).

$$\text{Relative Importance} = \frac{\text{Absolute Importance } (PS_j)}{\sum_{j=1}^k \text{Absolute Importance } (PS_j)} \quad (4)$$

where PS_j is the *j*th parameter ($1 \leq j \leq n$) and *n* is total number of predictors.

Next, models with five different numbers of inputs (predictors $n = 27, 10, 3, 2, 1$) are discussed.

4.1. Twenty-seven inputs

To reflect the importance of parameters used for building the power prediction model and their past states, the initial value *m* of the past states is set to 5 (50 s = 5 past values measured at 10 s intervals). Thus, the total number of predictors (input states) *n* equals 27 (5 WS past states + 5 PO past states + 5 RS past states + 5 GT past states + 5 BPA past states + 1 current GT state + 1 current BPA state). The input space is represented as vector (5).

$$IS_{27} = [v(t-1), \dots, v(t-5), x_1(t), \dots, x_1(t-5), x_2(t), \dots, x_2(t-5), y_1(t-1), \dots, y_1(t-5), y_2(t-1), \dots, y_2(t-5)] \quad (5)$$

An NN algorithm is applied to determine the importance of predictors. The first ten most important predictors among all the 27 states in Eq. (5) are shown in Table 4.

As illustrated in Table 4, the ten most important predictors amount to 97.76% of the total importance score of all 27 states. To reduce the computational effort, the remaining 17 states contribute 2.34% of the overall importance and are not considered for further analysis.

4.2. Ten inputs

Here, $n = 10$ parameters listed in Table 4 are used as inputs to the NN model. The input space is represented as the vector of predictors (6).

$$IS_{10} = [y_1(t-1), \dots, y_1(t-3), y_1(t-5), y_2(t-1), \dots, y_2(t-4), x_2(t), x_2(t-1)] \quad (6)$$

The importance of predictors is computed by the NN algorithm as shown in Table 5.

The sequence of predictors in Table 4 differs with Table 5, yet GT (*t*) and RS(*t* − 1) remain on the top of both lists.

4.3. One, two, and three inputs

For $n = 3$, the input space (the parameters highlighted in bold in Table 5) is represented as the vector of predictors (7).

$$IS_3 = [y_1(t-1), y_2(t-1), x_2(t)] \quad (7)$$

The importance of predictors recomputed with an NN algorithm is shown in Table 6.

Table 7Prediction results for different number of predictors n .

Number of predictors	data set	Average observed power output [kW]	Average predicted power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
27	Training	442.75	442.74	8.48	7.11	4.25	20.08
	Test	<i>605.56</i>	<i>605.56</i>	<i>10.72</i>	<i>9.01</i>	<i>2.20</i>	<i>5.84</i>
10	Training	442.75	442.73	8.60	7.43	4.04	14.44
	Test	605.56	605.34	10.78	9.56	2.23	7.14
3	Training	442.75	442.68	8.83	7.65	4.89	62.14
	Test	605.56	605.17	10.98	9.78	2.25	5.73
2	Training	442.75	442.74	8.86	7.66	4.33	19.84
	Test	605.56	605.38	10.97	9.77	2.26	6.50
1	Training	442.75	442.66	9.42	7.91	4.48	16.54
	Test	605.56	604.88	11.40	9.89	2.34	5.20

Based on (12), the predictor vectors for $n = 2$ and $n = 1$ are presented in (8) and (9), respectively.

$$IS_2 = [y_2(t-1), x_2(t)] \quad (8)$$

$$IS_1 = [x_2(t)] \quad (9)$$

4.4. Comparison of prediction performance

To evaluate prediction performance of the models developed in this paper, the following metrics are used: the Mean Absolute Error (MAE), Mean Relative Error (MRE), Standard Deviation of MAE, and Standard Deviation of MRE [11,18].

To compare the performance of models with a different number of predictors (inputs), here $n = 27, 10, 3, 2, 1$. The prediction accuracy of both training and test data is summarized in Table 7. For each n , 30 NNs are constructed, and the best performing NN is selected for comparative study.

The best test prediction performance reported in Table 7 was attained for $n = 27$ predictors (the row in italics) for the four metrics (MAE, MRE, STD of MAE, and STD of MRE). A reduced number of predictors has decreased prediction accuracy for the test data. Compared with $n = 27$, for $n = 3$, the MAE decreases by 0.26 kW, and it decreases by 0.68 kW for $n = 1$.

5. Clustering input space

Computational experience with the five clustering scenarios is discussed in the next five sections.

5.1. Clustering on a single past state of wind speed

The generated wind power is impacted by the wind speed $v(t)$; therefore, here and in the scenario to follow, it is used to cluster the

input space. After the training data has been clustered on $v(t)$ with the k -means algorithm into k clusters, the corresponding value of the generator torque $x_1(t)$, rotor speed $y_2(t-1)$, and power output $y_1(t-1)$ are assigned to each instance in every cluster. Note that the wind speed $v(t)$ is used only for clustering, without being used to train an NN model. The value of wind speed $v(t)$ is estimated by two methods: one using one past state of the wind speed only, i.e., $v^*(t) = v(t-1)$, and the other (see Eq. (10)) in the form of a time series model.

For the training data set clustered on wind speed $v(t-1)$ by the k -means algorithm and number of clusters, $k = 2, \dots, 5$. After the best performing NN has been selected (out of 30 NNs produced for each cluster), the prediction results for the test data are shown in Table 8.

As illustrated in Table 8, the best prediction accuracy is attained for $k = 4$ (shown in bold). However, it is not significantly better than the base performance for vector space (7) with MAE = 10.98 kW, STD of MAE = 9.78 kW, MRE = 2.25%, and STD of MRE = 5.73% (see in bold in Table 7).

5.2. Clustering on the wind speed estimated by a time series model

Using a time series model is another way to estimate the wind speed $v(t)$ based on the model in (10).

$$v(t) \approx v^*(t) = f(v(t-1), \dots, v(t-5)) \quad (10)$$

Here, the five past values of wind speed $v(t-1), \dots, v(t-5)$ serve as inputs to an NN algorithm. The estimated wind speed $v^*(t)$ is used for clustering by the k -means algorithm, for $k = 2, \dots, 5$. The corresponding values of the generator torque $x_1(t)$, rotor speed $y_2(t-1)$, and power output $y_1(t-1)$ are used to build 30 NN models for each cluster. The best performing NN model is selected, and the prediction results for the test data are shown in Table 9.

Table 8Prediction performance based on the data clustered on the past state of wind speed $v(t-1)$.

Input space clustered by	Number of clusters	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
WS($t-1$)	2	605.56	605.29	10.90	9.64	2.29	8.63
WS($t-1$)	3	605.56	605.22	10.98	10.45	2.29	7.90
WS($t-1$)	4	605.56	605.32	10.86	9.54	2.27	6.04
WS($t-1$)	5	605.56	605.40	10.89	9.69	2.33	10.80

Table 9Prediction performance based on the data clustered on the wind speed $v^*(t)$ of (17).

Input space clustered by	Number of clusters	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
WS*(t)	2	605.56	605.27	10.99	10.41	2.25	4.87
WS*(t)	3	605.56	605.14	11.00	9.90	2.38	12.80
WS*(t)	4	605.56	605.40	11.03	10.62	2.38	11.39
WS*(t)	5	605.56	605.05	11.27	10.44	2.39	8.97

Table 10
Prediction performance based on the data clustered on generator torque $x_2(t)$.

Input space clustered by	Number of clusters	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
GT(t)	2	605.56	605.29	10.74	9.44	2.23	6.56
GT(t)	3	605.54	605.37	10.79	9.50	2.23	5.96
GT(t)	4	605.56	605.14	10.58	9.11	2.16	5.48
GT(t)	5	605.56	605.25	10.58	9.24	2.15	4.47

Table 11
Prediction performance for clustering by GT (t) and RS ($t - 1$).

Input space clustered by	Number of clusters	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
(GT(t), RS($t - 1$))	2	605.56	605.36	10.83	9.79	2.22	5.80
(GT(t), RS($t - 1$))	3	605.56	604.10	11.17	10.07	2.25	5.13
(GT(t), RS($t - 1$))	4	605.56	605.05	10.78	9.70	2.21	5.76
(GT(t), RS($t - 1$))	5	605.56	605.10	10.80	9.63	2.25	6.51

The best prediction accuracy attained by the NN algorithm for $k = 2$ is shown in bold in Table 9. However, this prediction accuracy is worse than the base performance for the vector space (7) reported in bold in Table 7.

Though the wind speed directly contributes to the power output generation, clustering input space by the wind speed does not benefit prediction accuracy. This might be because the current wind speed $v(t)$ is not known, and both estimations carry error impacting the prediction accuracy.

5.3. Clustering on generator torque $x_2(t)$

Generator torque $x_2(t)$ is the most significant predictor, with an importance of 94.28% among all 27 predictors (see Table 4). Therefore, it is used to cluster the input space, with the corresponding rotor speed $y_2(t - 1)$ and the power output $y_1(t - 1)$ assigned to each cluster. The test data is categorized according to the nearest cluster centroid. The performance of the best NN models (out of 30 for each cluster) is shown in Table 10.

The best prediction accuracy is attained for $k = 4$, as shown in bold in Table 10. This prediction accuracy is significantly improved compared with the vector space of Eq. (7) (see in bold in Table 7).

5.4. Clustering on generator torque $x_2(t)$ and rotor speed $y_2(t - 1)$

Generator torque $x_2(t)$ and rotor speed $y_2(t - 1)$ have a combined importance of 95.67% (94.28% + 1.39%) of all the

provided 27 predictors (see Table 4). In this scenario, they are used to cluster the input space with the corresponding power output $y_1(t - 1)$ assigned to each cluster. The test data is categorized according to the nearest cluster centroid. The performance of the best NN model (out of 30 for each cluster) is shown in Table 11.

The best prediction accuracy occurs for $k = 4$, as shown in bold in Table 11. This prediction accuracy is significantly improved over the vector space of Eq. (7) (see the bold entries in Table 7).

5.5. Clustering on generator torque $x_2(t)$, rotor speed $y_2(t - 1)$, and power output $y_1(t - 1)$

Generator torque $x_2(t)$, rotor speed $y_2(t - 1)$, and the power output $y_1(t - 1)$ have a cumulative importance of 96.13% (94.28% + 1.39% + 0.46%) of all 27 predictors (see Table 4). In this scenario, they are used to cluster the input space. Each test instance is labeled with the nearest cluster centroid. The number of clusters varies from two to five. The performance of the best NN models (out of 30 for each cluster) is shown in Table 12.

The best prediction accuracy for $k = 4$ is shown in bold in Table 12.

5.6. Description of the clustered input space

The most promising scenario for clustering the input space is based on the generator torque $x_2(t)$. The number of clusters $k = 4$

Table 12
Prediction performance for clustering on GT(t), RS($t - 1$), and PO($t - 1$).

Input space clustered by	Number of clusters	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
(GT(t), RS($t - 1$), PO($t - 1$))	2	605.56	605.10	10.83	9.65	2.24	6.74
(GT(t), RS($t - 1$), PO($t - 1$))	3	605.56	605.04	10.73	9.30	2.16	4.71
(GT(t), RS($t - 1$), PO($t - 1$))	4	605.56	605.30	10.72	9.33	2.19	5.17
(GT(t), RS($t - 1$), PO($t - 1$))	5	605.56	605.22	10.76	9.55	2.21	5.69

Table 13
Clustering centroids for training data set.

Cluster no.	GT(t)	PO(t)	PO($t - 1$)	RS($t - 1$)	Average distance to the centroid	Number of instances	Instance percentage (%)
1	13.89	112.84	115.13	12.17	0.05	8534	28.12
2	31.36	342.67	344.72	14.67	0.04	11 993	39.51
3	48.28	658.93	665.12	17.99	0.06	6683	22.02
4	84.56	1258.93	1240.31	19.78	0.09	3143	10.35

Table 14

Clustering centroids for the test data set.

Cluster no.	GT(<i>t</i>)	PO(<i>t</i>)	PO(<i>t</i> – 1)	RS(<i>t</i> – 1)	Average distance to centroid	Number of instances	Instance percentage (%)
1	17.55	148.87	164.4	12.4	0.05	715	4.51
2	32.21	357.51	362.6	14.93	0.04	6107	38.51
3	49.95	693.19	706.28	18.32	0.05	6910	43.57
4	79.79	1186.54	1132.69	19.72	0.09	2127	13.41

Table 15

Relative distances between cluster centroids for four predictors.

Cluster No.	GT(<i>t</i>)	PO(<i>t</i>)	PO(<i>t</i> – 1)	RS(<i>t</i> – 1)
1	26.35	31.93	42.80	1.89
2	2.71	4.33	5.19	1.77
3	3.46	5.20	6.19	1.83
4	5.64	5.75	8.68	0.30

produces the best prediction accuracy among $k = 2, \dots, 5$. The centroids for the training data sets are shown in Table 13.

Four clusters of the training data are illustrated in Table 13. The generator torque $x_2(t)$ is used to cluster input space by k -means

where PS_j is the j th parameter state, $1 \leq j \leq 3$; N_i is the number of instances of the i th cluster, $1 \leq i \leq 4$.

Table 14 illustrates four cluster centroids of the test data set.

Four clusters of the test data set are illustrated in Table 14 for the parameters of Eq. (7), the observed power output $y_1(t)$, average distance to centroids, and the number of instances in each cluster. The similarity between training and test clusters is measured by the relative distance between the cluster centroids of the training and test data for each parameter state. A shorter relative distance between cluster centroids implies greater similarity between them. The relative distance metric is defined in (12):

$$\text{Relative Distance} = \frac{|\text{Training Cluster Centroid} - \text{Test Cluster Centroid}|}{\text{Training Cluster Centroid}} \times 100\% \quad (12)$$

algorithm. The clustering results represent four levels of the generator torque $x_2(t)$ from low to high with the corresponding observed power output $y_1(t)$. Clusters are arranged by the observed power output $y_1(t)$ from low to high, e.g., cluster 1 contains data with the lowest generator torque $x_2(t)$ and lowest observed power output $y_1(t)$, while cluster 4 includes data with the highest values of the generator torque $x_2(t)$ and the observed power output $y_1(t)$.

Instances of test data are categorized according to the cluster centroids of the training data using the algorithm in Table 3. After all instances are categorized, the cluster centroids of the test data are measured by average values through each cluster for each parameter state (see equation (11)).

$$\text{TestClusterCentroid}_{ij} = \frac{\sum_{i=1}^{N_i} PS_j}{N_i} \quad (11)$$

Table 16

Test performance of five data-mining algorithms for cluster 1 data.

Algorithm	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
RFA	148.87	148.40	9.62	9.84	6.46	88.75
BTA	148.87	148.86	6.12	4.77	5.51	11.54
SVM	148.87	164.40	6.48	4.93	4.35	24.46
NN (Ensemble)	148.87	148.63	6.25	4.95	4.20	25.62
NN	148.87	148.57	6.21	5.06	5.88	24.21

Table 17

Test performance of the five data-mining algorithms for cluster 2 data.

Algorithm	Average observed power output [kW]	Average predicted power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
RFA	357.51	356.86	24.40	22.39	6.45	5.82
BTA	357.51	357.53	11.16	10.07	3.25	2.96
SVM	357.51	362.60	9.75	7.92	2.83	2.28
NN (Ensemble)	357.51	357.29	8.98	7.70	2.59	2.17
NN	357.51	357.24	8.94	7.62	2.58	2.14

Table 18

Test performance of the five data-mining algorithms for cluster 3 data.

Algorithm	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
RFA	693.19	693.60	38.79	43.69	5.22	5.57
BT	693.19	692.96	12.04	10.48	1.76	1.58
SVM	693.19	706.28	13.12	10.40	1.95	1.62
NN (Ensemble)	693.19	692.42	11.87	10.22	1.75	1.54
NN	693.19	692.35	11.82	10.14	1.74	1.52

Table 19

Test performance of the five data-mining algorithms for cluster 4 data.

Algorithm	Observed average power output [kW]	Predicted average power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
RFA	1186.54	1185.95	21.87	20.99	1.82	1.66
BT	1186.54	1186.25	13.96	11.75	1.19	1.02
SVM	1186.54	1132.69	14.07	11.03	1.21	0.99
NN (Ensemble)	1186.54	1187.14	12.90	10.77	1.11	0.96
NN	1186.54	1187.03	12.93	10.77	1.11	0.96

Ensemble. The test performance of each algorithm for cluster 1 (of Table 14) is shown in Table 16.

The average observed power output for cluster 1 in Table 16 is 148.87 kW. The previous research has shown that predicting power output at low levels with an NN resulted in large errors, e. g., when the observed power output was less than 10 kW, the relative error was as high as 300%, or even 1000%. Note that the errors reported in Table 16 for low values of power are smaller.

Of all the algorithms in Table 16, the Boosting Tree Algorithm (BTA) outperformed both the NN and NN (Ensemble) on the four metrics (MAE, MRE, STD of MAE, and STD of MRE), as indicated in bold in Table 16. Therefore the BTA is selected for constructing the model based on cluster 1 data. The learning rate of $\alpha = 0.3$ has produced the best prediction accuracy.

6.2. Cluster two

The test performance for the five data-mining algorithms for cluster 2 data (see Table 14) is shown in Table 17.

As illustrated in Table 17, the average observed power output for cluster 2 is 357.51 kW. The best performance (see in bold) is attained by the NN selected of 30 NNs.

6.3. Cluster three

The test results of the five data-mining algorithms for cluster 3 data (see Table 14) are shown in Table 18.

The average observed power output of cluster 3 is 693.19 kW (Table 18). The best prediction accuracy is achieved by the NN algorithm (the best one among 30 NNs tested).

Table 20

Best performing clusters.

Cluster No.	Average observed power output [kW]	Average predicted power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE	Algorithm
Cluster 1	148.87	148.86	6.12	4.77	5.51	11.54	BT
Cluster 2	357.51	357.24	8.94	7.62	2.58	2.14	NN
Cluster 3	693.19	692.35	11.82	10.14	1.74	1.52	NN
Cluster 4	1186.54	1187.14	12.90	10.77	1.11	0.96	NN (Ensemble)
Average	605.56	605.14	10.57	9.08	2.14	3.08	

6.4. Cluster four

The test performance of the five data-mining algorithms on cluster 4 data (see Table 14) is provided in Table 19.

Table 19 shows the average observed power output of 1186.54 kW. The best performance has been produced by the NN Ensemble algorithm with 5 (out of 30) best performing NNs.

6.5. Overall results

The best prediction accuracy results for the four clusters are summarized in Table 20.

The average prediction accuracy reported in Table 20 has been significantly improved compared with the performance based on the space vector in Eq. (7), with MAE = 10.98 kW, STD of MAE = 9.78 kW, MRE = 2.25 kW, and STD of MRE = 5.73% (see in bold in Table 7).

7. Comparative study

7.1.1. Comparison of four typical scenarios

Four typical scenarios from Sections 4, 5, and 6 are compared.

Scenario A

3 model inputs (generator torque $x_2(t)$, rotor speed $y_2(t-1)$ and power output $y_1(t-1)$) are used by the NN model. The best performing NN out of 30 NNs is selected.

Table 21
Comparison of test results.

Scenario	Average observed power output [kW]	Average predicted power output [kW]	Mean absolute error [kW]	STD of MAE	Mean relative error [%]	STD of MRE
A	605.56	605.17	10.98	9.78	2.25	5.73
B	605.56	605.56	10.72	9.01	2.20	5.84
C	605.56	605.14	10.58	9.11	2.16	5.48
D	605.56	605.14	10.57	9.08	2.14	3.08

Scenario B

27 model inputs (see Eq. (2)) are used by an NN model. The best performing NN out of 30 NNs is selected.

Scenario C

3 model inputs (generator torque $x_2(t)$, rotor speed $y_2(t-1)$ and power output $y_1(t-1)$) are used by a NN model. Input space is clustered by the generator torque $x_2(t)$ with the k -means algorithm. An NN algorithm is used to extract the prediction model for each cluster. For each extraction, 30 NNs are produced, and the one with the best performance is selected.

Scenario D

3 model inputs (generator torque $x_2(t)$, rotor speed $y_2(t-1)$ and power output $y_1(t-1)$) are used by data-mining models. Input space is clustered by generator torque $x_2(t)$ using the k -means algorithm. For each cluster, an algorithm with the best prediction performance is selected to extract the model (see Table 20).

Table 21 lists the prediction accuracy results of test data for the four scenarios.

Scenario D produces the best prediction accuracy for most metrics (MAE, MRE, and STD of MRE) in Table 21. The STD of MAE ranks the second best, and it is only slightly worse (by 0.07 kW) than the ones produced in Scenario B.

8. Conclusion

Predicting future production of wind power at low wind speeds is challenging. Estimates of power at low levels are extremely inaccurate. This paper proposed a clustering-based method for power prediction at low wind speeds using 10 s data. Numerous data-mining algorithms were developed for low level power prediction, including neural networks (NNs).

The approach proposed in the paper takes advantage of data subspaces that lead to accurate predictive models. The successful results reported in the paper were accomplished in a number of steps. First, significant parameters were selected by physics-based equations and data-mining algorithms. Second, training and test data sets were clustered according to five different criteria (scenarios). The clustering algorithm used here was the k -means algorithm. Third, for the data in each cluster the most suitable algorithm for building a power prediction model was identified.

The computational results reported in the paper demonstrated that the proposed model customization approach produced accurate prediction models using a small number of input parameters.

The latter provided important side benefits, including reduced computational effort and increased reliability.

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